

DEVELOPMENT OF RADIAL MAGIC FINGER DESIGN FOR PERMANENT MAGNET QUADRUPOLE

T. Brookbank, S. Brooks, O. Chubar, F. DePaola, M. Musardo, P. N'Gotta, S. Sharma
NSLS-II, Brookhaven National Laboratory, Upton, NY, USA

Abstract

Permanent magnets (PM) have been widely used in synchrotron light sources for insertion devices and recently for the accelerator multipole magnets. The NSLS-II accelerator upgrade is based on the complex bend lattice that will use combined function permanent magnet-based dipole-Quadrupole (PMQ). The Halbach type design is considered for the PMQs. To achieve the required field quality with the Halbach PMQ, a field harmonic correction method based on assembly of small PM blocks called “Magic fingers” (MF) is developed. This paper presents the radial MF magnetic and mechanical design, the prototype and the correction results.

INTRODUCTION

The Halbach magnet design composed with PM segment can generate a strong multipolar magnetic field with a compact structure suitable for new generation of light sources. The future complex bend lattice, envisioned for NSLSII upgrade, requires combined function dipole-quadrupole magnets with a field strength of 130 T/m and 0.5 T for the quadrupole and dipole component, respectively. The magnetic field quality describing the deviation from the ideal field, is defined as the part of unwanted multipoles (harmonics), and specified at 10 units (the field main multipole component is normalized at 10000 units).

A Halbach PMQ prototype, using 16 PM wedges, was designed and built to achieve the combined function magnet. The magnetic simulation predicts a field harmonic error larger than 100 units for the real magnet due to tolerances of the PM characteristics (remanence, magnetization angle) and wedge assembly and geometrical errors. The magnetic measurement of the in-house PMQ prototype confirmed this prediction with a field error of 137 units [1, 2]. This report presents the design and prototyping of the MF system and the field correction of the PMQ prototype with MF that achieves a field quality close to 10 units.

MAGNETIC DESIGN

The Magic Finger field correction method is based on an assembly of small rectangular PM blocks placed at the longitudinal extremity of the PMQ magnet. The MF assembly generates an additional field with harmonics of the opposite sign to cancel the field harmonics of the magnet. To generate the required arbitrary correction field, the radial magnetization direction (up or down) and the radial length of the MF assembly are controlled. The PMQ and MF model are built with RADIA to simulate the magnetic field (Fig. 1) [3].

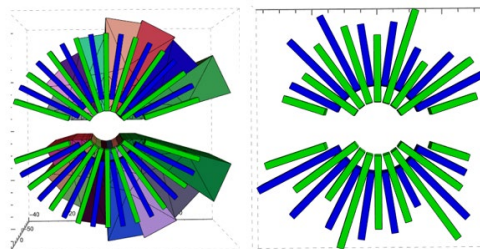


Figure 1: Halbach PMQ with MF correction.

The discrete MF rectangular PM block is a 2x2 mm² section and 15 mm longitudinal length. Two layers of PM stack (green, blue) are arranged to fully use the space. The harmonics correction capability of the MF is estimated using a statistical approach based on the generation of several random sets of MF assembly with random radial length and random magnetization direction. The generated harmonics strength is presented in Table 1.

Table 1: Maximum Harmonics Strength

Harmonic order	3	4	5	6	7	8
Strength [units]	215	75	21.5	10.5	3.8	0.5

The generated harmonics strength ranges from 215 units for the first order harmonic (n=3, sextupole), to lower value below 1 unit for the higher order (n≥8). This result gives the maximum strength for each harmonic when the MF assembly generates mainly a particular harmonic order.

Another simulation is performed with 1000 random MF assembly, and the harmonics root square sum (RSS) is calculated to estimate the MF correction capability. The results shown in Fig. 2, give an estimate of the MF correction capability of 150 units RMS, that is higher than the specified 100 units PMQ errors after assembly [1].

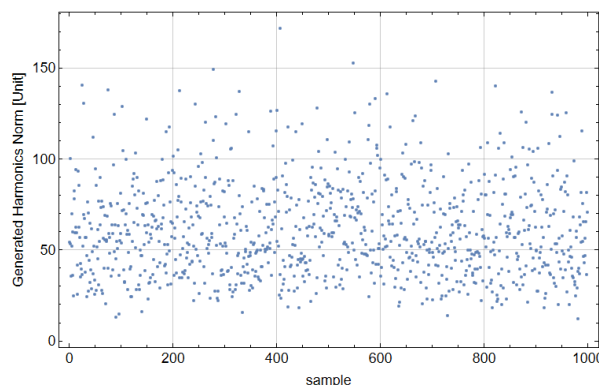


Figure 2: RSS of harmonics generated by 1000 random MF assembly.

MECHANICAL DESIGN

Constraints

The complex-bend lattice design posed mechanical challenges for the Magic Fingers due to the tight longitudinal spacing between the PMQs (≤ 15 mm). The required 100 units harmonics correction capability necessitates a MF length larger than 10 mm that is not a standard magnet dimension for MF design [4].

The MF container case design uses a circular array with a total of 38 slots, each accommodating rectangular PM blocks that are 2 mm wide, with discrete varying radial lengths of—2 mm, 3 mm, and 8 mm—to allow different combinations. NdFeB PM material (N35UH) with 1.2 T remanence and high coercivity of 2000 kA/m is selected for the MF.

The PMQ and vacuum chamber were finalized before MF design, limiting the available space (Fig. 3). Ideally, MF magnets would be at the same bore radius (8 mm) and occupy the full gap between the PMQ magnets. This would give the highest efficiency of the MF, but 0.5 mm of retaining material at the inner radius was required. The final minimum radial distance was 8.5 mm with a 12 mm magnet depth, allowing 1 mm of material on either side for the containment.

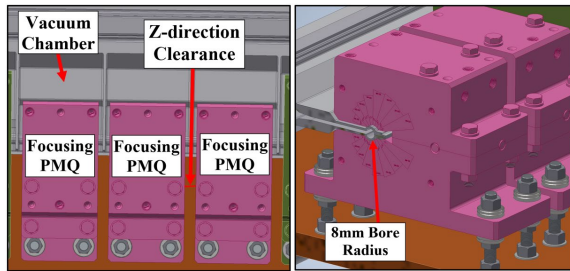


Figure 3: Focus PMQ's, clearance and bore radius.

To accommodate the clamshell PMQ structure, MF assemblies were split into top and bottom halves (Fig. 4), ensuring repeatable installation and alignment with the PMQ's mechanical center. Locating pins engage arc slots in the PMQ housing for concentric positioning. An angle adjustment ($\pm 1^\circ$) of the MF assembly is used to adjust the normal and skew components of the correction field harmonics using spring-loaded screws.

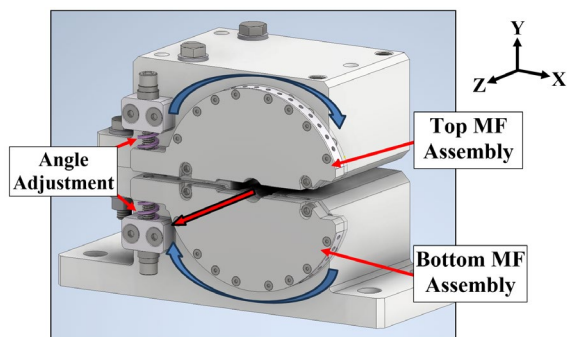


Figure 4: Magic Finger Assembly mounted to PMQ.

Based on magnetic simulation, each PMQ will have a tailored MF configuration for various field harmonics. Analysis of magnetic force showed that, even with all slots populated, Z-direction repulsion is acceptable, and transverse forces are negligible. The maximum magnetic force is 11.6 N in the Z-direction, Fig. 4. Despite variations in MF assembly, the forces remain well controlled, ensuring that the magnets can be safely installed into the assembly. Since the MF magnets are captured in the assembly, the MF container provides a mechanical advantage and doesn't cause a safety issue when assembling onto the PMQ.

Prototype

The first iteration of the prototype MF container was 3D printed in 316-stainless steel, Fig. 5(a). This allowed for fast prototyping and a design with complex features, Fig 5(b).

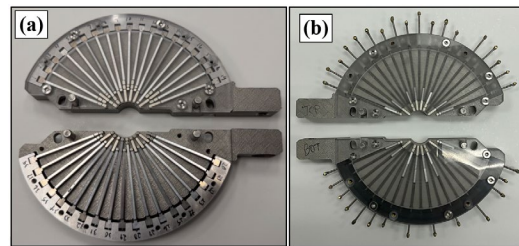


Figure 5: MF assemblies, (a) original with aluminum shims. (b) design change with threaded rods.

These prototypes allowed for quick testing of the design but are not suitable for mass production of the MF. A design that can use wire electrical discharge machining (EDM) would really benefit for mass production and assembly. A redesign after magnetic measurement to improve the design will be discussed.

MAGNETIC MEASUREMENT

With the PMQ's small aperture we utilized the in-house rotating coil (RC) bench to measure the PMQ before and after tuning. The RC bench uses a small diameter coil probe (12 mm) that fits the magnet aperture with a measurement radius of 5 mm (Fig. 6). The RC bench can measure the field harmonics up to the order 15 with a precision of 0.1 unit [5].

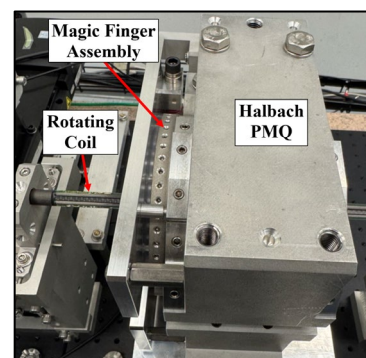


Figure 6: Assembly mounted to PMQ on RC Bench.

The field quality of the PMQ was expected to be within 100 units, given the tolerance on PM remanent field (Br) of 1%, magnetization angle error of 1.5 degree and assembly error of 100 μm .

The measured RSS of the harmonics was 137 units. Most of the contribution of RSS was given by the first harmonic order $n=3$ (sextupole). The RSS of the harmonics is larger than the expected value of 100 units due to large assembly errors of 800 μm for the top and bottom vertical gap [1]. Table 2 shows the normalized multipoles strengths.

Table 2: Normalized Multipoles Strengths of In-house PMQ.

Multipole Order n	Normalized b_n [Units]	Normalized a_n [Units]
1	12831.04	-0.1908
2	10000.00	0.0000
3	134.95	6.6436
4	25.41	-3.3420
5	7.03	-4.1287
6	-3.73	2.3839
7	1.3872	0.8363
8	0.5718	0.2361
RSS	137	

A MF assembly is calculated to generate the harmonics presented in Table 3 with the opposite sign, to cancel the field harmonics of the PMQ magnet. Two assembly solutions are presented in Figure 7. The first solution uses one MF layer (the closest to the bore) that is the most efficient and simplifies the assembly. The second solution uses the two MF layers. The assembly solutions are obtained using either Gauss-Newton method or a genetic algorithm [6,7].

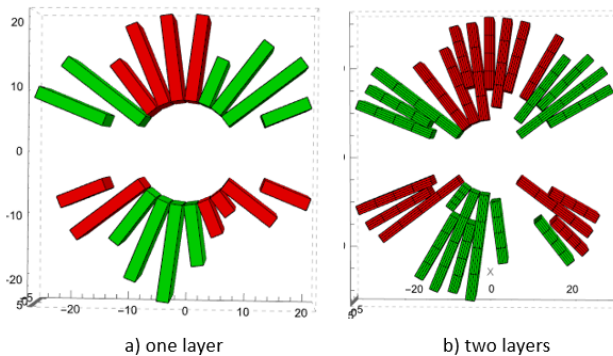


Figure 7: MF correction assembly solutions.

The magnets were assembled into the MF assembly according to each solution and measured on the RC bench. The harmonics correction results using these two different solutions are presented in Table 3.

Table 3: RSS Values for the Initial, One- and Two-layers

	Initial	One-Layer	Two-Layer
RSS	137	16.4	13.8

The results show that the one-layer and two-layer configurations give similar results, and both provide effective harmonics correction.

Although, the MF assembly was redesigned to utilize what was learned from the prototype and testing (Fig. 8). In the redesign, the second layer was removed, that makes manufacturing the MF container simpler. The angle adjustment was removed because it was found not useful. Also, removing the angle adjustment allowed the design to be made symmetrical. This allowed one design for top and bottom, reducing cost. These changes also allowed the MF container to be made simpler, which can now utilize wire EDM process.

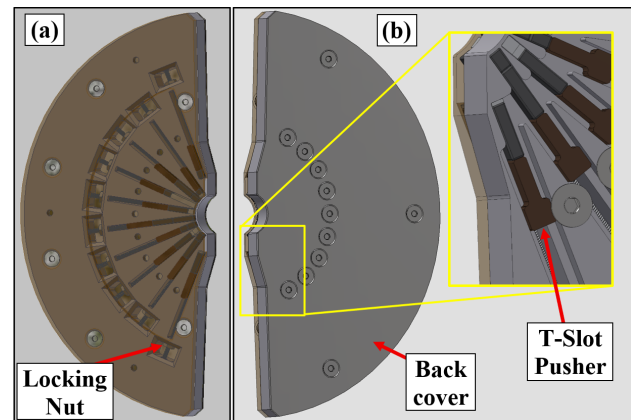


Figure 8: MF Assembly Redesign, (a) front view, (b) back view with detailed view.

The magnet clamping was changed to make installing the magnets easier and faster. The magnet clamping utilizes a socket head cap screw that pushes on a bronze T-slot pusher and can be locked with the locking nut. The shape of the pusher is important because the screw diameter is only 1.5 mm (needs to fit in the 2 mm slot) and can bend while tightening, so the T-slot stops the pusher from displacing in the slot direction. The screw tip also protrudes into the pusher block; this stops the screw from drifting up the bronze pusher, giving one degree of freedom to the screw. Overall, these design changes solve the issues that were found on the initial prototype.

CONCLUSION

The design and test of a MF harmonics correction for the Halbach PMQ magnet was presented. The MF correction test on the PMQ prototype with an initial harmonic RSS of 137 units resulted in a final harmonic RSS close to 10 units confirming the efficiency of this method. The MF container design was updated to simplify the design and make the MF assembly easier and faster.

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